Poster blocks 3 and 5 — Role B handoff

Sep 23, 2026 · @Someone

Both blocks are finished and inside the word budget. Two figures at 300 dpi, sized for one A0 column at 1:1 — drop them in at 100% and the point sizes below are what prints. Poster due Sep 26, 2026.

Block 3 — Data sources

Heading: Six sources, one vocabulary

Figure: `block3_sources.png` — 9.8 × 5.0 in at 300 dpi



Every count read from the downloaded file, not a dataset description page

Body copy, 58 words:

> **Seek Australia** — 3,204 Data Scientist ads, 2019–2022, full text: what the local market actually asks for. **ESCO** — 13,960 skill concepts with alt-labels: the shared vocabulary. **O*NET** — 8,753 tool names ESCO omits. **UTS Handbook** — 70 subjects across six courses, scraped with learning outcomes. **LinkedIn US** — 1,355 data roles, 2023–24, full text: the cross-market check. **2,484 resumes** build the student profile.

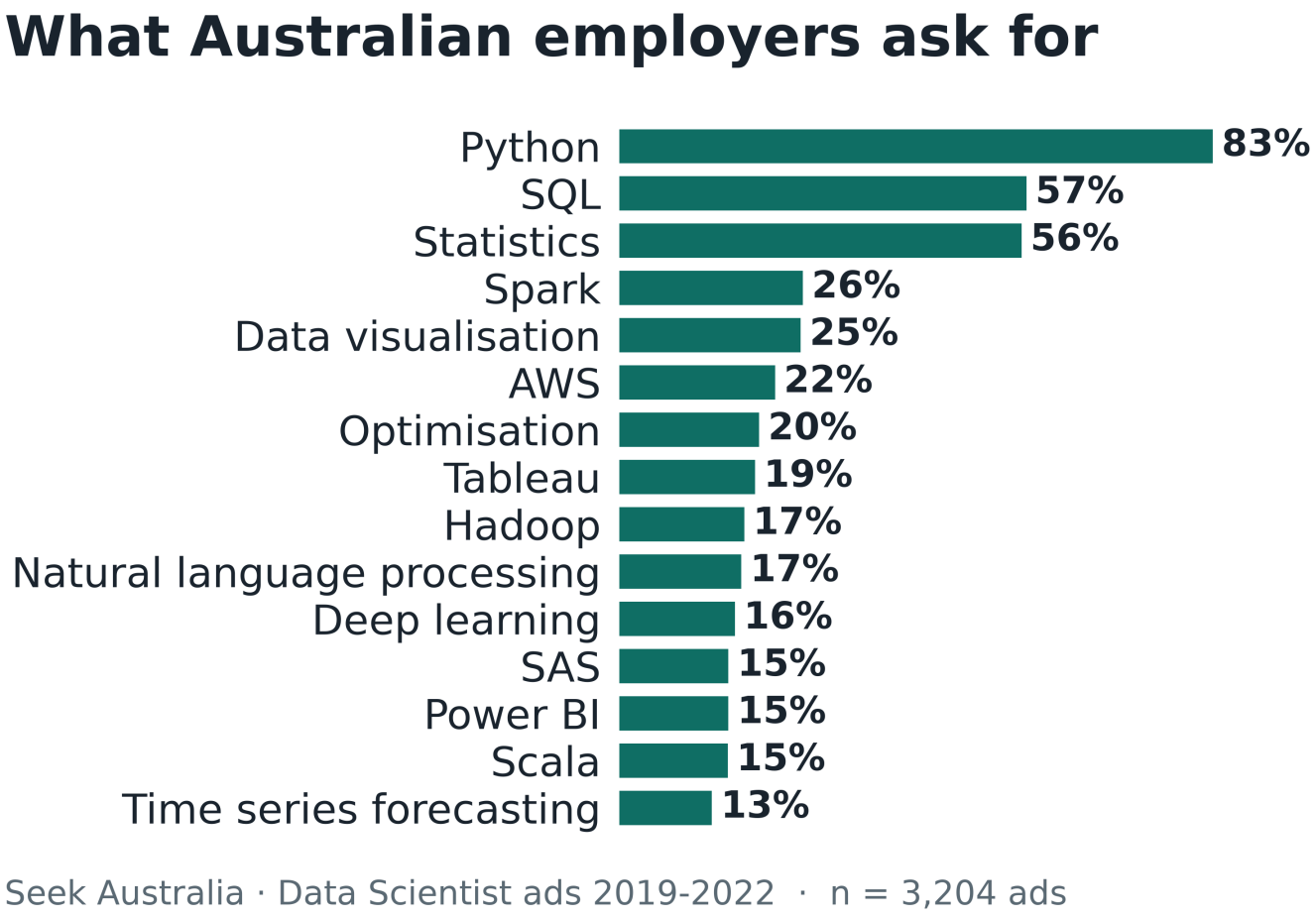
Caption: Every count read from the downloaded file, not a dataset description page.

Two sources named in the original plan changed. Seek could not be scraped — their terms forbid it — so the ads come from a published Seek dataset instead, which is the same data obtained legitimately. The handbook gave 70 subjects, not the ~180 the plan estimated, because that is all six courses actually list.

Block 5 — What the market asks for

Heading: What Australian employers ask for

Figure: `block5_top_skills.png` — 9.8 × 7.0 in at 300 dpi



Body copy, 40 words:

> Python leads at 83%, SQL 57%, statistics 56% — then a long tail where no tool passes 26%. The surprise sits off the chart: **communication appears in 28% of ads, more often than Tableau, AWS or Hadoop.**

Caption: Seek Australia · Data Scientist ads 2019–2022 · n = 3,204 · technical skills shown, professional skills discussed in text.

The chart follows the rules we agreed: two colours, direct value labels instead of an x-axis, no legend, every label horizontal, body text at 20 pt and above.

The US data earns its place — two findings

We ran the **same extractor** over LinkedIn's US ads, which is what makes this comparison legitimate. The 2023 Australian corpus could not be used this way because it ships pre-extracted skills and no ad text, so any difference would have measured the two extractors rather than the two markets.

1. The ranking is stable across markets. Spearman rank correlation **0.819** over 100 shared technical skills, 15 of the top 20 in common, and a mean gap of 4.9 points across the top 10.

Skill	Australia 2019–22	US 2023–24
Python	82.8%	85.4%
SQL	56.8%	59.5%
Spark	25.6%	24.3%
AWS	21.7%	23.9%
Tableau	18.9%	18.6%

That is a real validation result: two continents, two job boards, two time windows, and the core ranking holds. It justifies building on the larger Australian corpus.

2. It proves the recency gap rather than hiding it. Every LLM-era skill sits at exactly **0.0%** in the 2019–22 Australian ads and appears in the 2023–24 US ads: large language models 0 → **15.8%**, PyTorch 3.0 → 18.6%, Docker 2.7 → 10.9%, LangChain 0 → 1.6%, prompt engineering 0 → 1.2%, RAG 0 → 0.4%. Hadoop moves the other way, 17.4% → 10.5%.

So the missing LLM stack is a property of the sampling window, and we can now show that instead of merely asserting it. **One caveat to state:** geography and time change together between the two corpora, so a single difference cannot be attributed to either alone. The LLM result is safe because those tools did not exist commercially before 2022.

Two things the poster must say

The ads are 2019–2022. No LLM, RAG or prompt engineering appears in the chart. We can now show this is the sampling window rather than the Australian market, because those skills sit at 0.0% in the 2019–22 ads and at 15.8% in the 2023–24 US ads. Say it in the caption, with the number — a marker who spots the gap unprompted will read it as an oversight.

Professional skills are off the chart on purpose. Communication, teamwork and leadership rank 4th, 7th and 11th overall. They are real, not noise. They are excluded because they do not map to subjects, which is what the product does, and the caption states the choice rather than hiding it.

Numbers the other blocks can use

For C, block 7 — a worked example of the whole product. AWS appears in 22% of these ads. No MDSI subject mentions AWS or cloud at all. Master of AI has two: Architecting on Amazon Web Services (42049) and Cloud Computing and SaaS (42904). Both are open to an MDSI student as free electives, and neither is visible from their own course page. Market skill → measured curriculum gap → the subject that fills it. Worth spot-checking by hand before it goes up, since it is keyword matching over subject text.

For D, block 8 — a measured baseline. Exact label matching maps 36% of our 133-skill dictionary onto ESCO. 73 skills, 55%, have no ESCO concept at all: Docker, Kubernetes, XGBoost, Snowflake, Tableau and the entire LLM stack. That is the number a semantic matcher has to beat, and it is why the traditional-versus-neural comparison is worth running.

Files. Everything is in `data-eda/` — figures in `outputs/`, the data manifest in `DATASETS.md`, the scripts in `src/`. Every count traces to a file on disk.